

CREDIT SCORING MODEL

Machine Learning Project Report

Objective: Predict the creditworthiness of an individual using past financial and credit-history data.

Problem Type: Supervised Learning – Binary Classification

1. Introduction

Credit scoring estimates whether a borrower is likely to be a good or bad credit risk. This project demonstrates a complete machine-learning workflow using financial features and two classification algorithms: Logistic Regression and Random Forest.

2. Dataset

The project uses a reproducible synthetic dataset of 1,500 customer records. Synthetic data is used for academic demonstration and does not contain real personal financial information.

Feature	Meaning
Age	Customer age
Income	Customer income
Debt	Existing debt
Loan Amount	Requested loan amount
Employment Years	Years of employment
Credit History Years	Length of credit history
Number of Loans	Number of previous/current loans
Late Payments	Count of late payments
Payment History Score	Payment reliability score
Credit Utilization	Approximate utilization ratio
Credit Status	Target: Good Credit / Bad Credit
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3. Methodology

The data is divided into 80% training and 20% testing sets. Numeric missing values are handled with median imputation and numerical features are standardized. Class weighting is used to reduce the effect of class imbalance. Logistic Regression provides a baseline classifier, while Random Forest captures nonlinear relationships.

Workflow: Dataset → Preprocessing → Train/Test Split → Model Training → Prediction → Evaluation

4. Evaluation Metrics

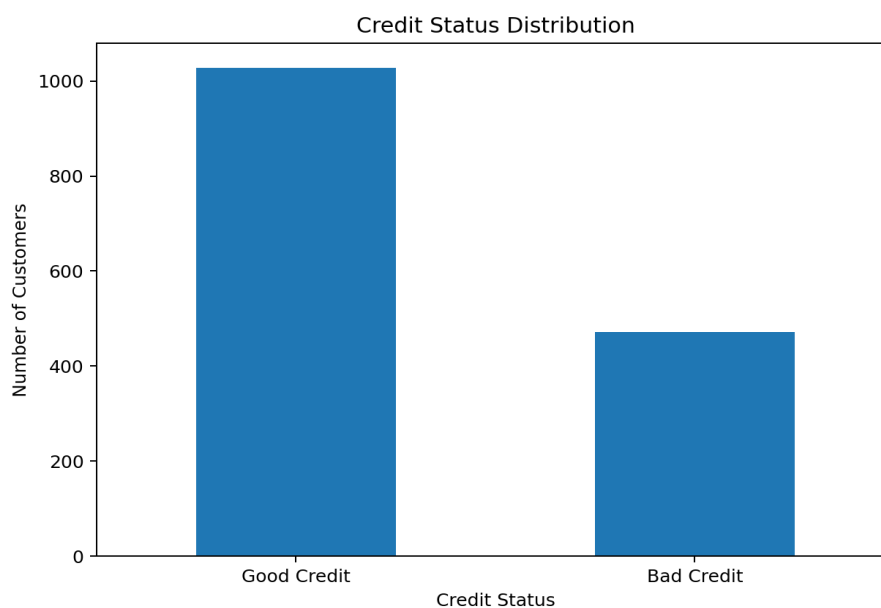
Accuracy measures overall correctness. Precision measures the correctness of positive predictions. Recall measures how many actual positive cases are detected. F1-Score balances precision and recall. ROC-AUC measures the ability to distinguish between the two classes.

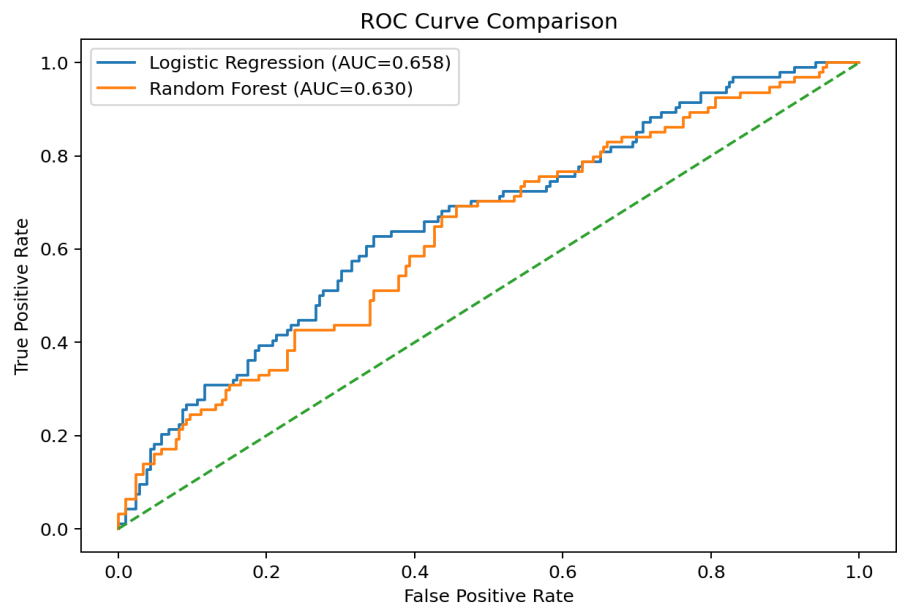
5. Model Results

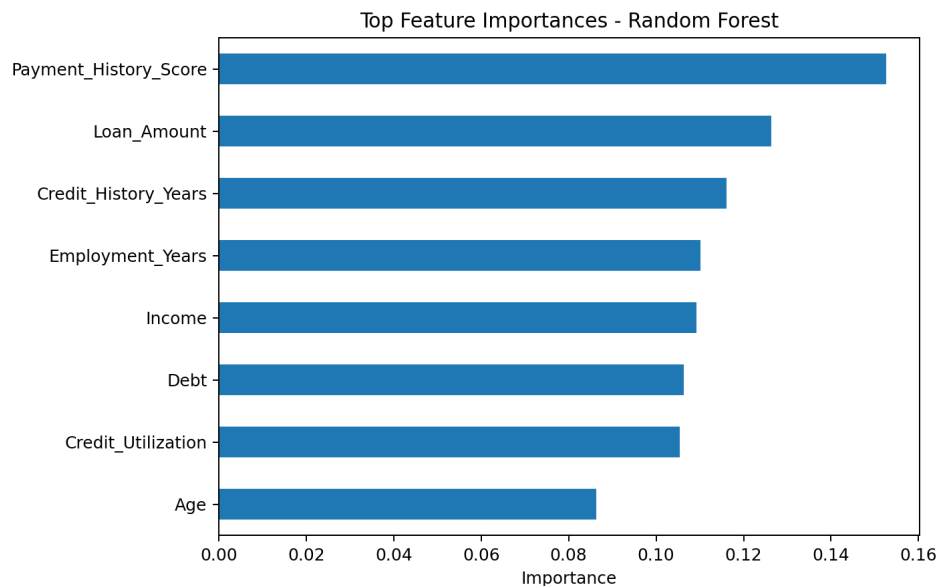
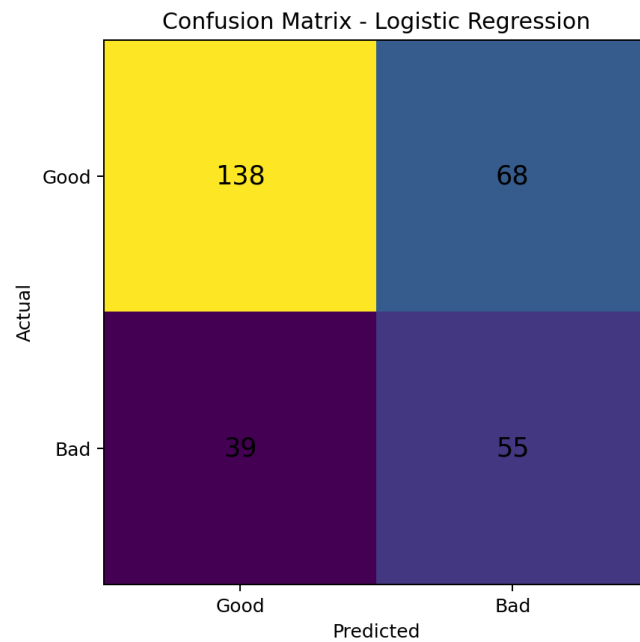
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.643	0.447	0.585	0.507	0.658
Random Forest	0.650	0.427	0.340	0.379	0.630

Best model by F1-Score: Logistic Regression

6. Visual Analysis







7. Conclusion

The project successfully demonstrates a credit-scoring classification pipeline. Logistic Regression achieved the highest F1-Score on the included synthetic test set. Using several evaluation metrics gives a more complete picture than accuracy alone. Before real-world deployment, the model would require validated data, fairness analysis, explainability, security controls, regulatory review, and independent validation.

8. Future Scope

- Use a validated real-world credit dataset.
- Tune hyperparameters with cross-validation.
- Investigate class imbalance and threshold selection.
- Add model explainability.
- Evaluate fairness across relevant customer groups.
- Deploy through a secure web/API application.